AC CHARGER CONFIGURATION

V3.2

1. INTRODUCTION

New version is designed to make the configuration on webpage via PC, PAD and Mobile phone, no need extra configuration tool.

2. INTERNET CONNECTION

The AC Charger is supporting internet connection via Ethernet, Wi-Fi and 4G.

2.1 Ethernet Connection

Connect the Ethernet cable to Ethernet port of main board

2.2 Cellular Connection

Insert the SIM card to the 4G module

3. CONFIGURATION

It is recommended to use the Chrome kernel browser for access

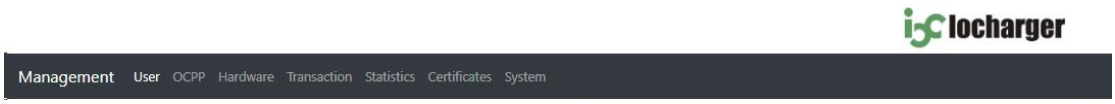
3.1 Configuration Webpage

Power on the charger and connect your PC, PAD or Mobile phone to the Charger Wi-Fi, SSID will be 'IOC-XXXXXX' series, password 'IOC12345'.

Login the webpage by entering the default IP address '192.168.10.1:8900' and enter Username 'admin' and Password 'ioc12345' then click 'ok'

3.2 DESCRIPTION

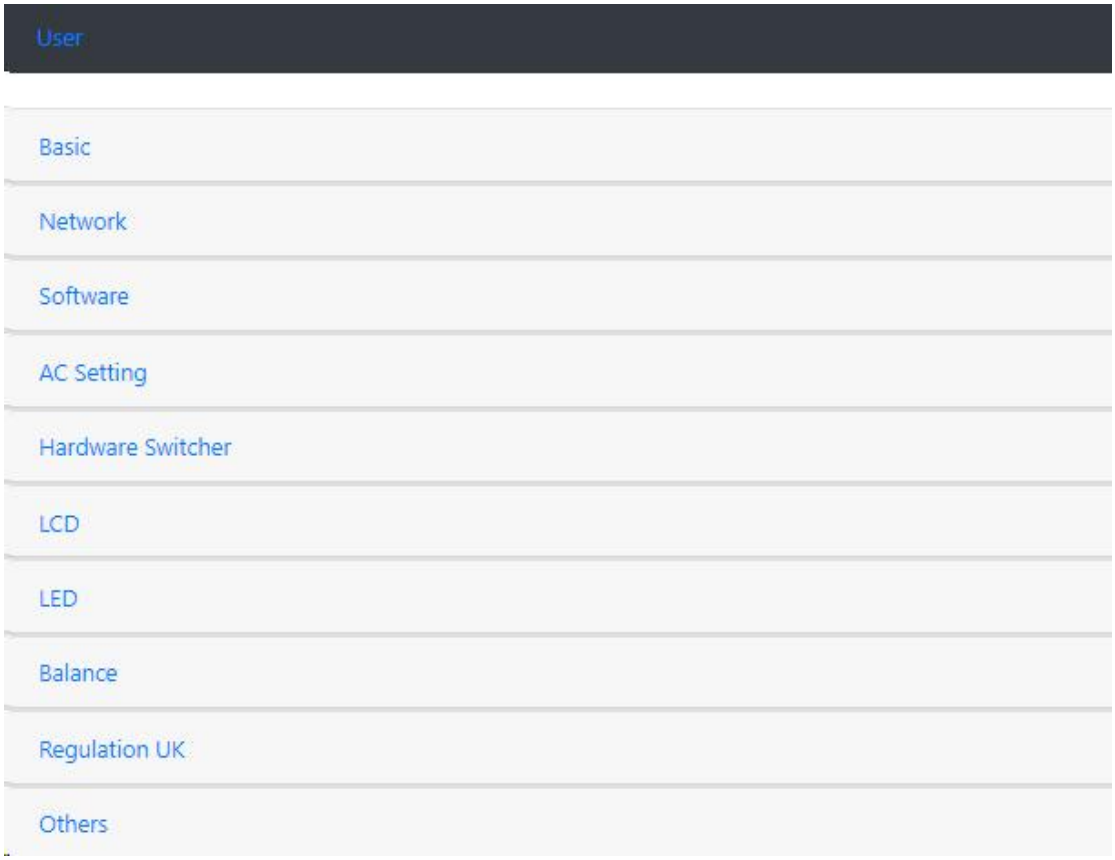
There are 7 pages available with the configuration page.



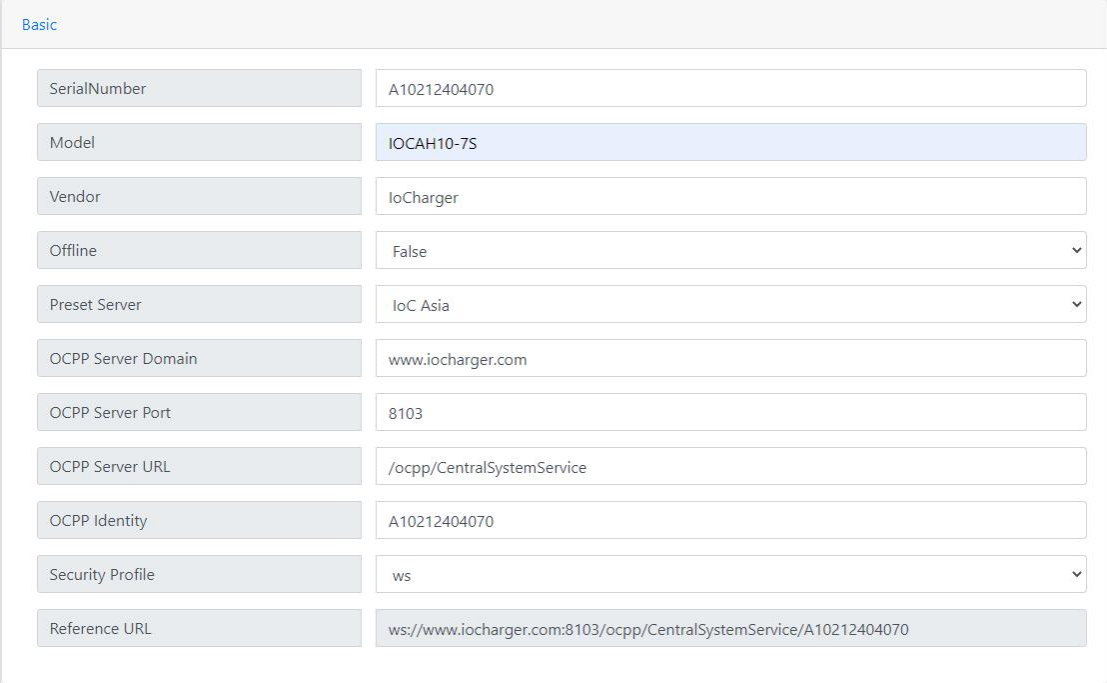
- User
  EV Charger basic configuration
- OCPP
  Standard OCPP2.0.1 configuration
- Hardware
  Hardware configuration, which is suggested not to be modified
- Transaction
  Transaction data of charging sessions
- Statistics
  Charger statistics for analysis
- Certificates
  Valid certificate installation
- System
  System settings

3.3 User

9 modules included, Basic, Network, Software, AC Setting, Hardware Switcher, LCD, LED, Balance, Regulation UK and Others.



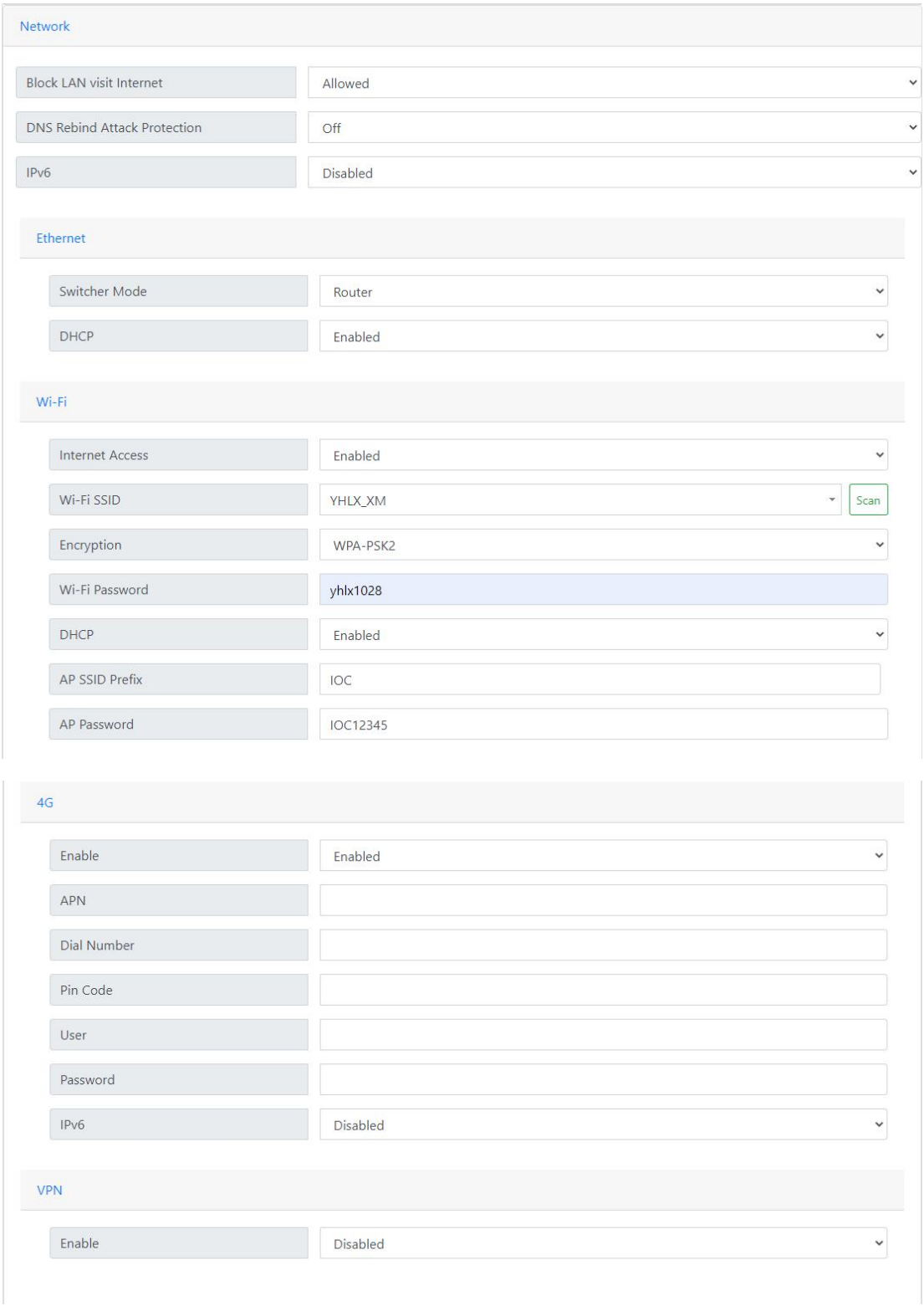
3.3.1 User Basic



- SerialNumber: Serial Number used for OCPP BootNotification to the server, and same as the sticker.
- Model: Charger Model used for OCPP BootNotification to the server.
- Vendor: Charger Vendor used for OCPP BootNotification to the server.
- Offline mode activated once the value is true, which will be not connecting to the OCPP server.
- Preset Server: Select the existing OCPP server or another user defined
- OCPP Server Domain: Domain used for OCPP server connection.
- OCPP Server Port: Server port used for OCPP server connection.
- OCPP Server URL: OCPP Server URL for OCPP connection.
- OCPP Identity: The unique ChargePoint ID assigned by the OCPP server.
- Security Profile: WS or WSS connection for OCPP server connection.
- Reference URL: Charger OCPP URL generated automatically

3.3.2 Network configuration

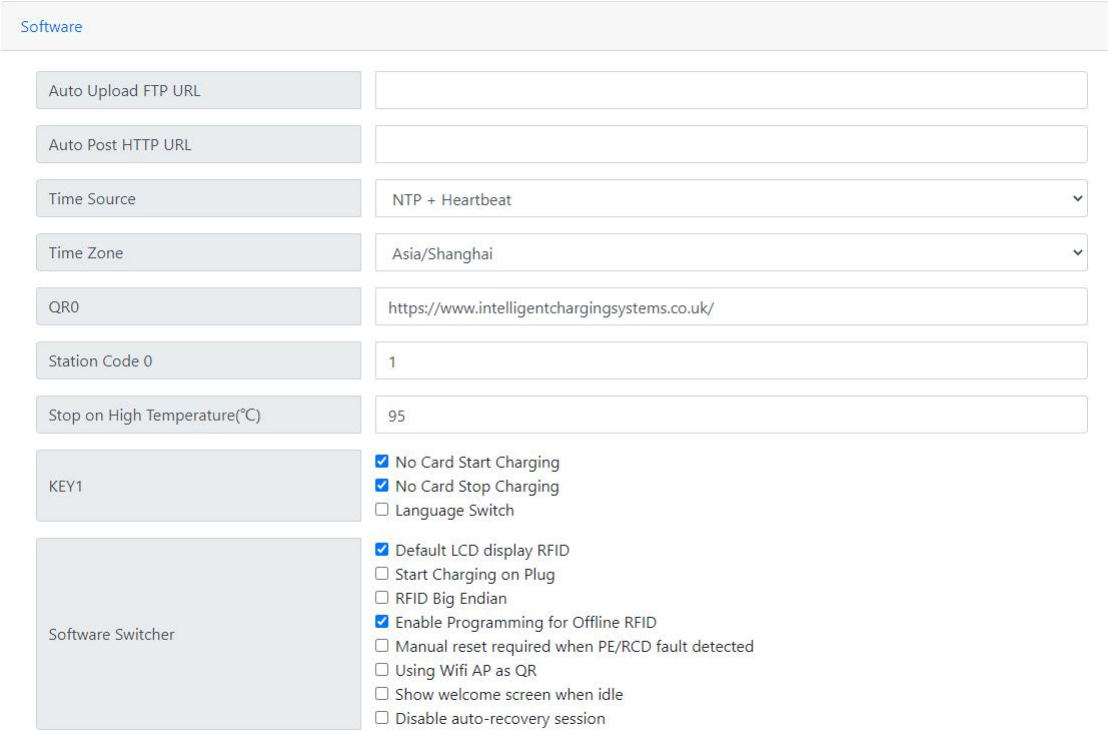
Network configuration: 5 options available, Block LAN visit Internet, Ethernet, Wi-Fi, 4G and VPN.



- Block LAN visit internet: User can visit internet when the value is false
- DNS Rebind Attack Protection
- IPv6: Activate it to allow user to access the charger via IPv6 address function

- Ethernet connection is available with both DHCP and static IP.
- Switcher Mode: only used for commercial charger for now, PC accesses the charger webpage configuration page via Ethernet cable. 3 options available, Router, Switcher and Cascade.
- Wi-Fi: Scan the Wi-Fi list and select the SSID to connect .
- 4G: Please check with the cellular provider about the cellular connection related information.
- VPN: Default disable and PPTP optional

3.3.3 Software Configuration

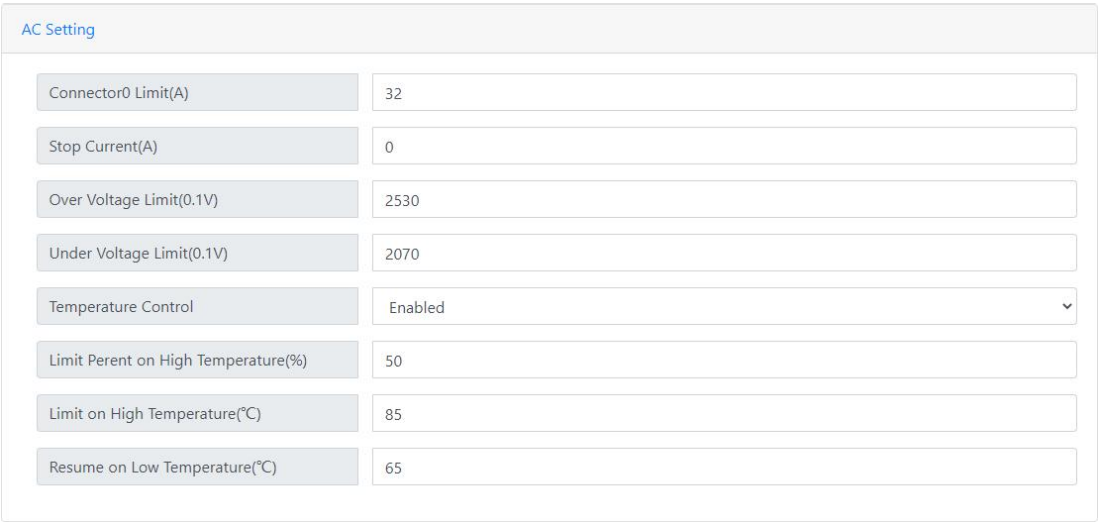


- Auto Upload FtpURL: Fill in server address and the charger will automatically upload the log at UTC 12:00 everyday
- Auto Post HTTP URL: No value required by default
- Time Source: default NTP+Heartbeat and NTP only, Heartbeat only optional
- Time Zone: Select the time zone for the charger configuration
- QR: Configure QR code to be showed on the screen display, QR0 for single Wallbox and QR0 & QR1 for pedestal dual AC Charger
- Station Code: Configure the Station code to be showed on the screen display, either ‘0’ or 4 digital numbers, Station Code 0 for single Wallbox and Station Code 0 & Station Code 1 for pedestal dual AC Charger
- Stop on High Temperature: Stop charging while reaching high temperature value.
- KEY: Select the option to activate the silver button function, KEY1 for single Wallbox and KEY1 & KEY2 for pedestal dual AC Charger

- No Card Start Charging, press the button to start charging
- No Card Stop Charging, press the button to stop charging
- Language Switch, press the button to switch screen language

- Software Switcher: Select the option to activate the software function
  - Default LCD display RFID, screen default display swipe the RFID card
  - Start Charging on Plug, just plug and charge, no need extra authorization.
  - RFID Big Endian, default will be small Endian, and select the option to activate the Big Endian
  - Enable Programming for Offline RFID, activate the offline RFID function, and swipe the RFID card while pressing the silver button to activate, and swipe the RFID card again while pressing the silver button to deactivate.
  - Manual reset not required when PE/RCD fault detected
  - Using Wifi AP as QR, scan the screen QR to recognize charger WiFi information function
  - Show welcome screen when Idle, screen will show welcome when charger is in idle status
  - Disable auto-recovery session, disable automatic recovery charging function after the power supply of the charger is powered off and then powered on again

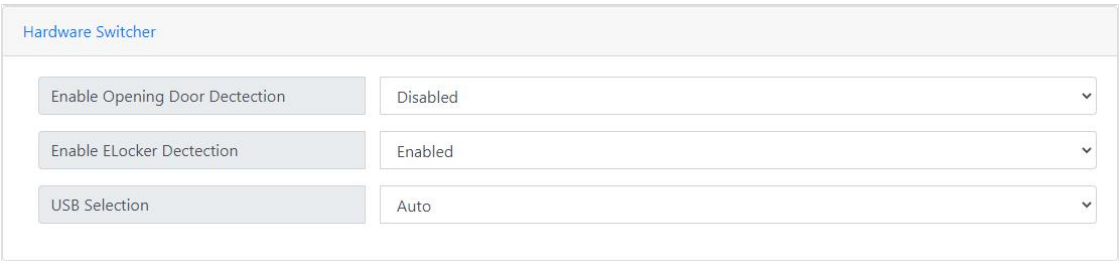
3.3.4 AC Setting



- Current Limit: Max output current
- Stop Current: Current to stop charging, usually set ‘0’ to be controlled from EV side
- Over Voltage Limit: Value for over voltage limit
- Under Voltage Limit: Value for under voltage limit
- Temperature Control: Recommend Enabled
- Limit Perent on High Temperature: Power limit on high temperature
- Limit on High Temperature: Temperature to limit power

- Resume on Low Temperature: Temperature to resume power

3.3.5 Hardware Switcher

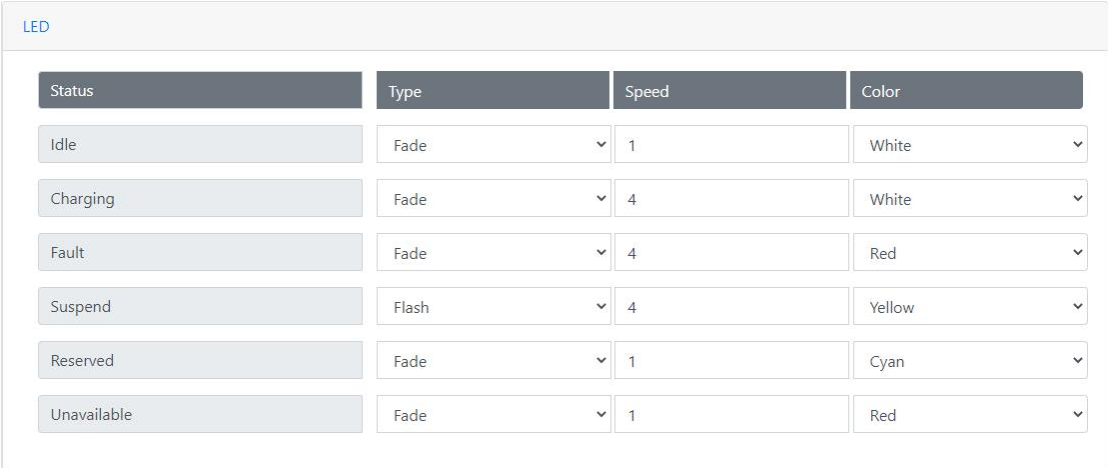


- Enable Opening Door Detection, need hardware support, not available as default
- Enable Elocker Detection: Activate the Elocker when the value is Enabled
- USB Selection: Default Auto and USB, 4G optional

3.3.6 LCD

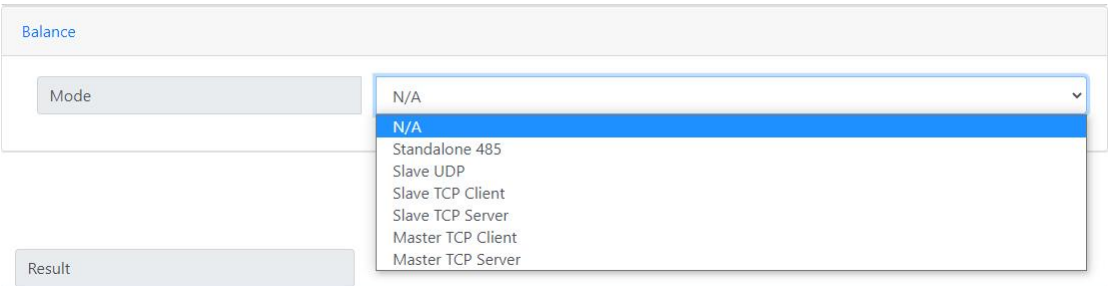
- Idle Brightness: Brightness of charger on Idle mode
- Active Brightness: Brightness of charger on active mode
- Charging Brightness: Specific brightness of charger during charging
- Active Timeout(minutes)
- Language: Configure the charger screen language, only multilingual firmware is supported

3.3.7 LED



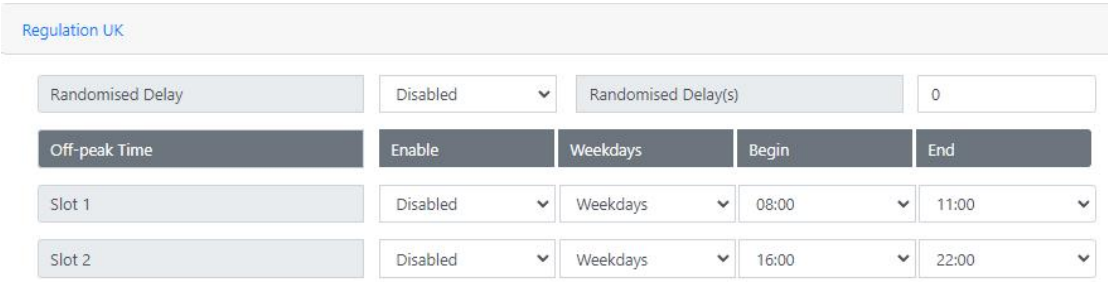
- Configure the LED display of charger on different status

3.3.8 Balance



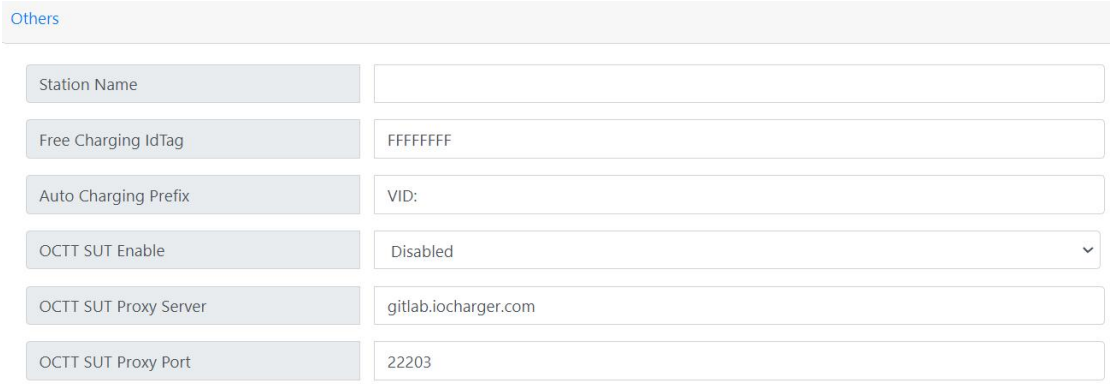
There are 7 models available for testing now, N/A, Master TCP Server and Master TCP Client, Slave TCP Server and Slave TCP Client, Slave UDP and Standalone 485, pls refer to the document " *Iochargre New Load Balancing Commissioning* " .

3.3.9 Regulation UK



- Configure the charging time of the charging station according to UK regulations.

3.3.10 Others



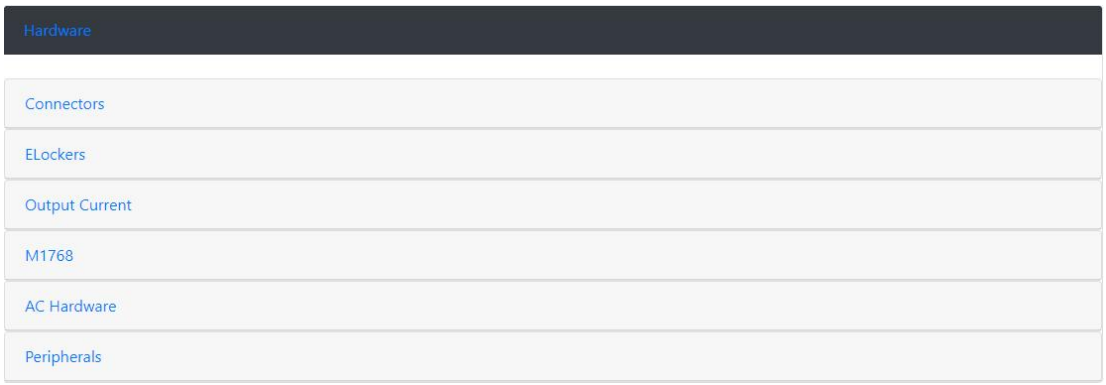
- Configure additional features.

3.4 OCPP

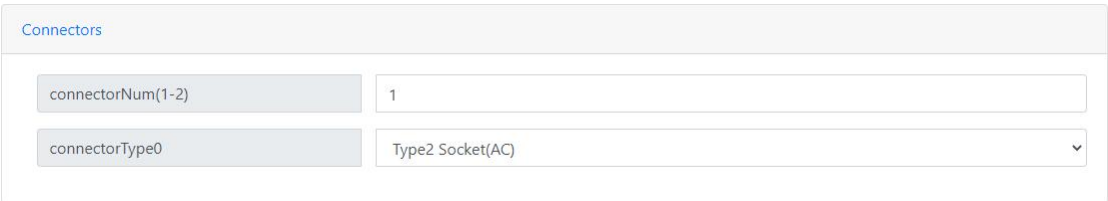
Standard OCPP Read/Write configuration, please refer to OCPP document for detailed explanation.

3.5 Hardware

Current charger hardware configuration, and may not modify the parameters without notice.



3.5.1 Connectors

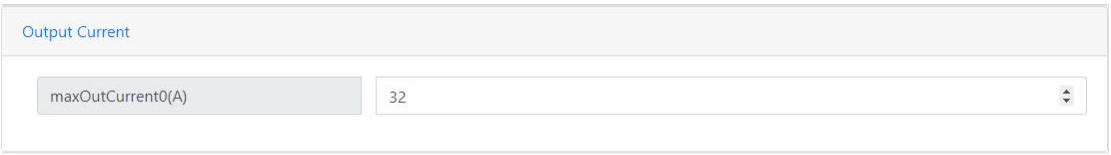


- ConnectorNum, configure the charging connector numbers
- Configure the Connector Type, Connector Type 0 for single Wallbox, and Connector Type 0 & Connector Type 1 for dual AC Chargers.

3.5.2 Connectors

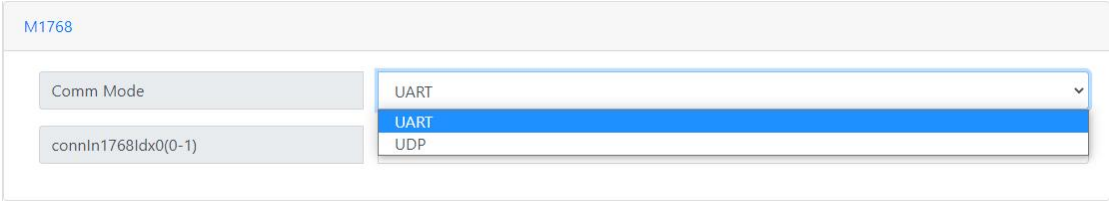
- Configure the Elocker type of charging connectors.

3.5.3 Output Current



- Configure the Max output current of charging connectors.

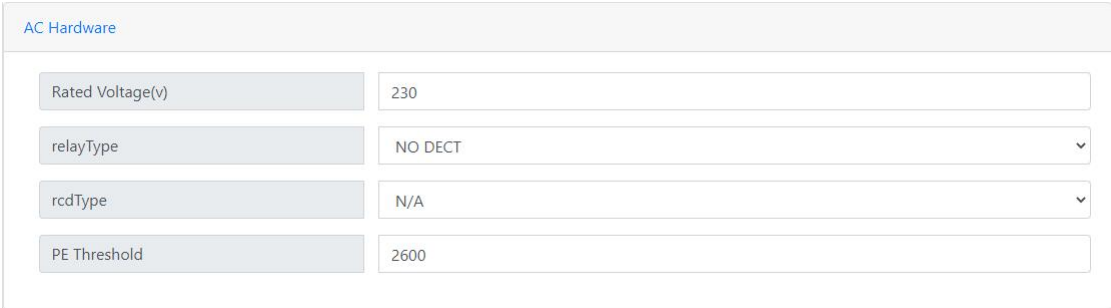
3.5.4 M1768, Main board data communication protocols



- UART for new case home charger,
- UDP for commercial chargers,

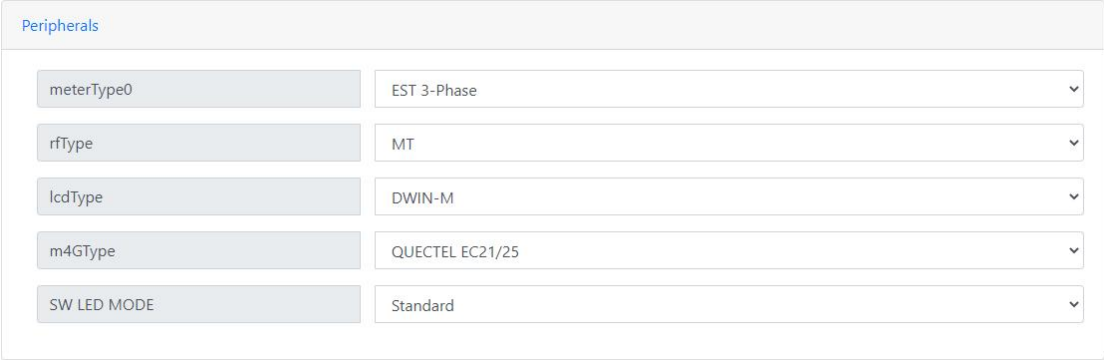
3.5.5 AC Hardware

- Configure the rated voltage, relay type and rcdType



- Need hardware support to monitor the relay

3.5.6 Peripherals



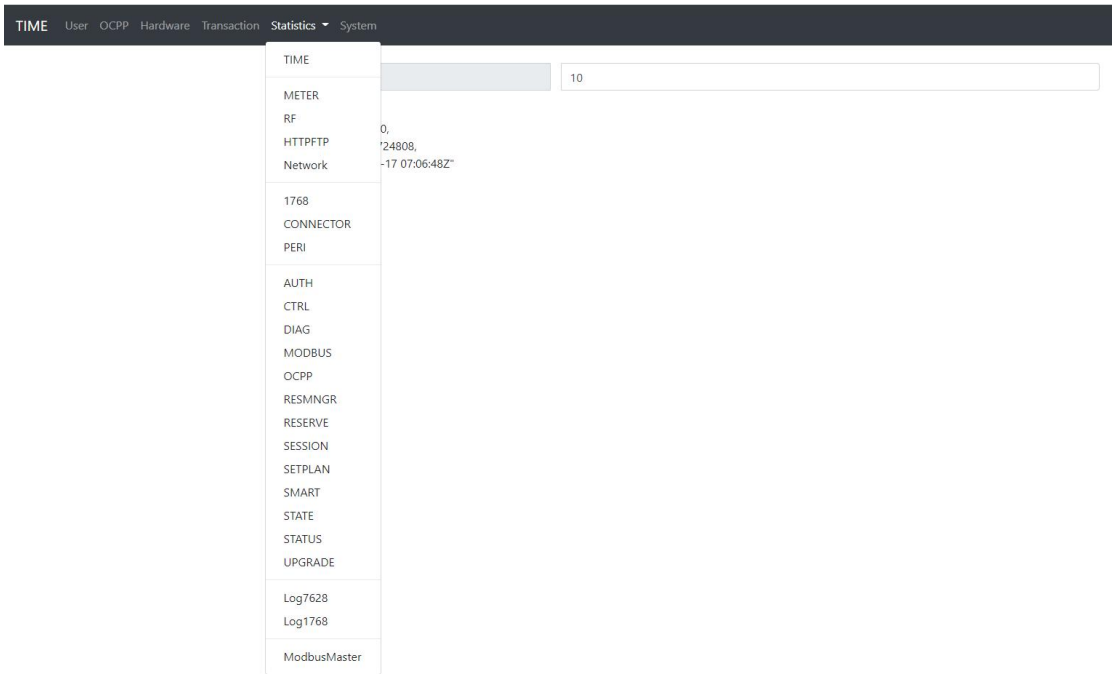
- Configure the meter type, RFID card reader type, LED type, LCD Type, and also 4G module type.

3.6 Transaction

This page contains the transaction data of charging session such as the time, fee, information of charger etc. for further analysis.

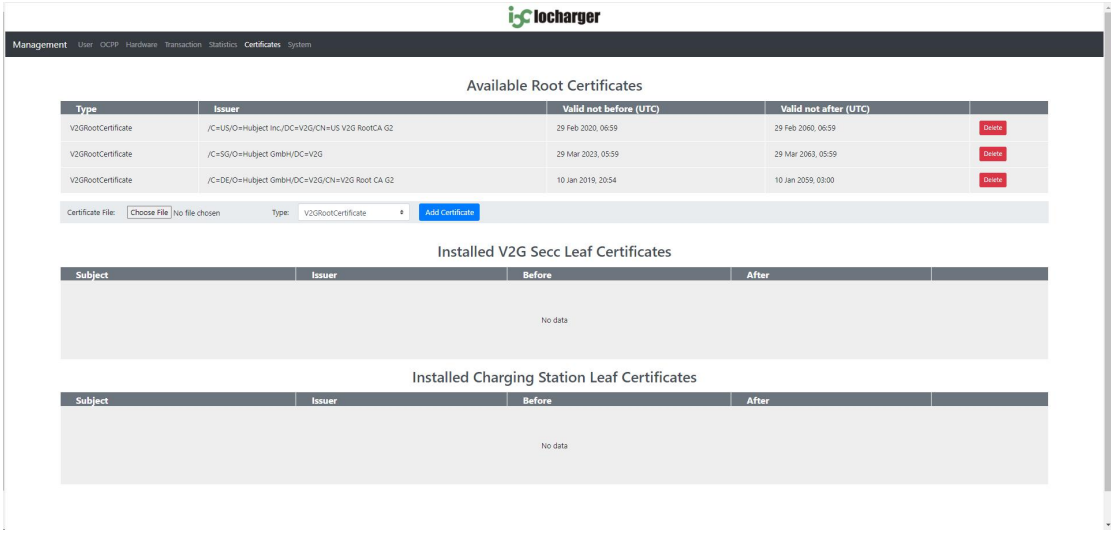
3.7 Statistics

This page contains the debugging configuration information of each module generated when the charger is running. It is used for the operator's administrator to collect all kinds of information on this page to assist IoCharger engineers in troubleshooting.



3.8 Cetificates

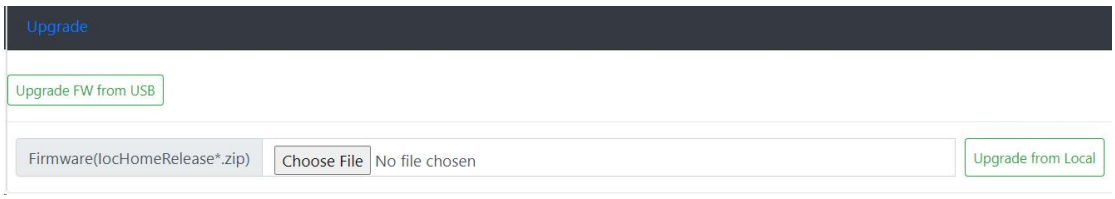
This page is about certificate installation. You can select a local certificate file and install it on the charging station. The types of certificates that can be uploaded include V2G root certificates, MO root certificates, CSMS root certificates, and manufacturer root certificates.



3.9 System

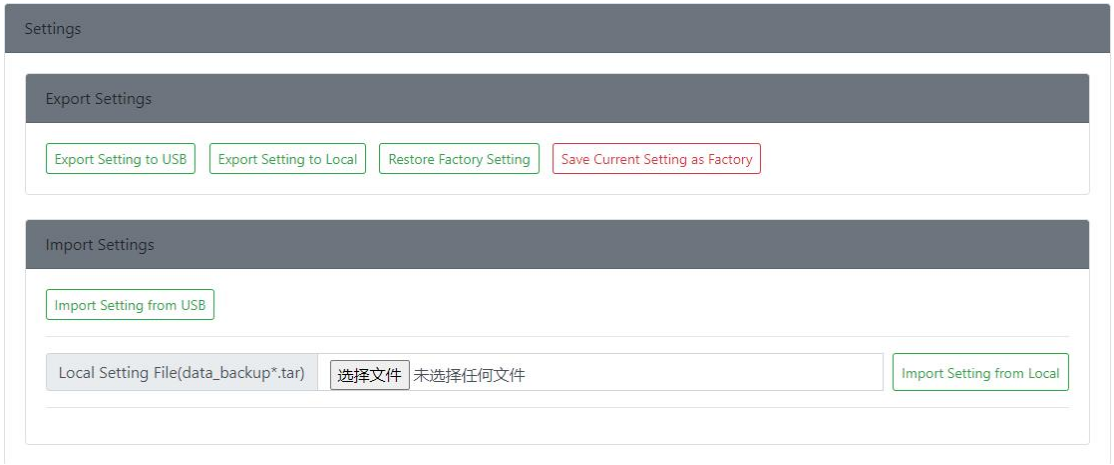
5 modules available, Upgrade, Settings, Diagnostics, Utility and ChangePassword.

3.9.1 Upgrade



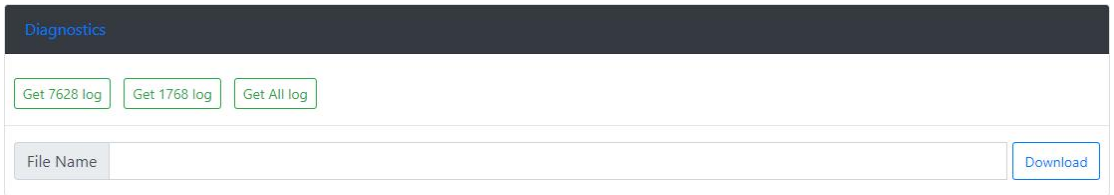
- Upgrade from USB: Upgrade via pan drive
- Upgrade from Local: Upgrade locally

3.9.2 Settings



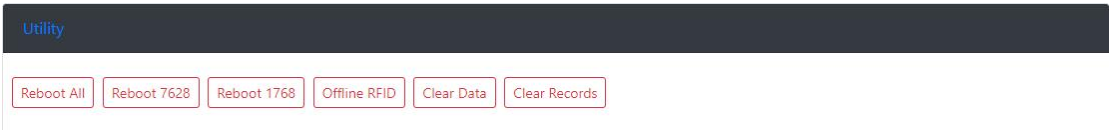
- Export and Import settings from USB or Local
- Restore Factory Setting: Complete reset
- Save Current Setting as Factory

3.9.3 Diagnostics



- Download the system log files to analyze fault reason.

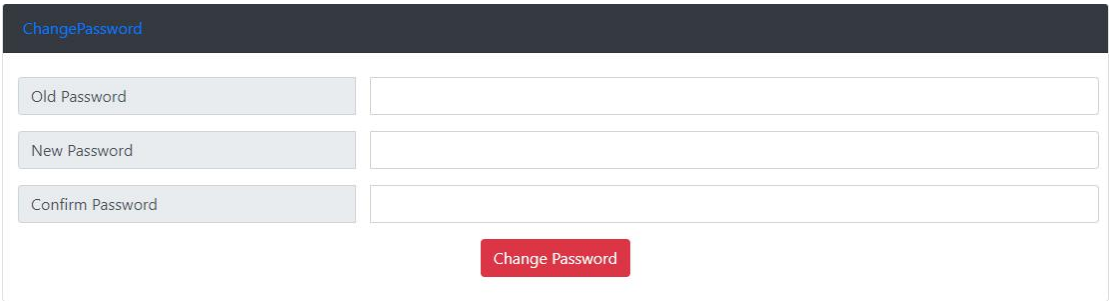
3.9.4 Utility



- Reboot All: Reboot the complete charger
- Reboot 7628: Reboot the complete charger 7628 board
- Reboot 1768: Reboot the complete charger 1768 board
- Offline RFID: Click the button to configure the offline ID, same function as silver button.
- Clear Data: Clear cache data of the charger
- Clear Records: Clear the charging records of the charger

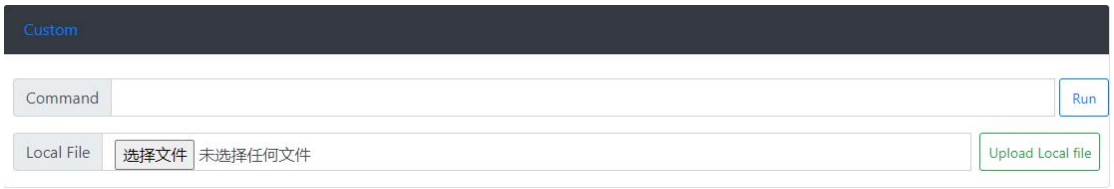
All these features are related to OCPP 2.0.1 testing. Please refer to the relevant OCPP documentation for operation

3.9.5 ChangePassword



- Change charger webpage configuration login password. Please do not change it at will or be sure to remember the new password.

3.8.6 Custom



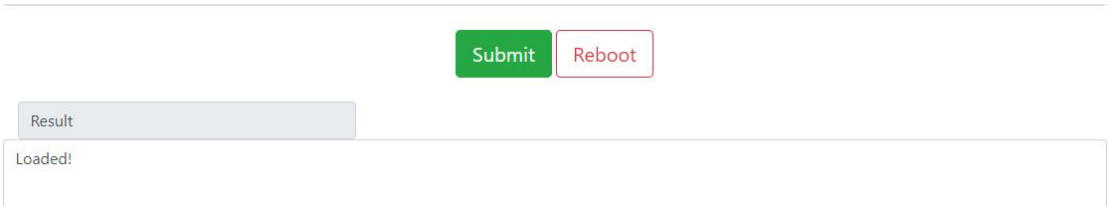
- Load customer's other commands to set or test the system.

3.10 Common Questions

3.9.1 Upgrade Notice

- Any new firmware upgrading proceeded, please remember to clear the browser caches.

3.9.2 Submitting a new configuration



- New configurations will be successfully loaded when the ‘Result’ value is “Loaded!” and the “Submit” button’s color is green.

3.9.3 Submitting fault

- Please modify the configuration parameters according to the prompt messages showed in the ‘Result’ table,
- Please contact IoCharger for any further technical support.

3.9.4 Because continuous improvement is a goal at iocharger, we reserve the right to make file modifications at any time.